



UNIVERSITY OF NAIROBI
DEPARTMENT OF CIVIL AND CONSTRUCTION
SCI 3266: COMPUTER SCIENCE III: FEB 2025
Lecturer : Mogire, omogire@uonbi.ac.ke, 0721441924

DEPARTMENT OF CIVIL AND CONSTRUCTION
COURSE OUTLINE

Code & Name	SCI 3266 COMPUTER SCIENCE III
Prerequisite	SCI 3265 COMPUTER SCIENCE II
Cohort	
Lecturer	Obwaya Mogire
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Purpose

Equip the learner with skills to choose correct algorithms and data structures in software development.

Course Objectives

1. Learn the basic data types used in computer science
2. Learn the strengths and weakness of various data types
3. Implement some abstract data types
4. Learn the concept of algorithm analysis
5. Apply the various analysis methods to optimize algorithms

References:

1. *Data Structures for Engineers and Scientists Using Python* by Rakesh Nayak and Nishu Gupta, 2025, CRC Press is an imprint of Taylor & Francis Group, LLC
2. *Data Structures Using C*, Reema Thareja, OXFORD

Internet material

1. <https://www.tutorialspoint.com/python/index.htm>
3. https://www.tutorialspoint.com/data_structures_algorithms/index.htm

Introduction

- Basic Definitions
- Structured Data Types
- Arrays

Lists

- Lists as an Abstract Data Type
- Implementation

Stacks

- Stack as an Abstract Data Type
- An Array Implementation of Stacks
- Application of Stacks

Queues

- Queue as an Abstract Data Type
- An Array Implementation of Queues
- Applications of Queues

Trees

- Binary Trees
- Binary Search Trees
- Heaps

Graphs

- Definition
- Representation

Lecture Hours: : 3 hours

- Traversing
- Minimum spanning tree
- Topological sort
- Shortest Path

Algorithms

- Analysis
- Performance Evaluation
- Asymptotic notations

Sorting Algorithms

- Straight Selection Sort
- Selection Sort
- Bubble Sort
- Insertion Sort
- Quick Sort
- Merge Sort
- Heap Sort

Searching algorithms

- Sequential Search
- Binary Search

Course Assessment

Laboratory work	5 %
Project:	5 %
Assignments:	5 %
CATs:	15 %
Final Exam:	70 %

Grading System

≥ 70 A

60 && 69 B

50&& 59 C

40 && 49 D

< 40 FAIL

Approved by: _____ Signature: _____ Date: _____